

1202SCG Calculus 1
1020ENG Engineering Mathematics 2

Week 9 Lecture

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Integration by Parts: A Repeating Integral

Evaluate

$$I = \int e^x \cos x \, dx.$$

Choose $u = \cos x$ and $v' = e^x$. Then $u' = -\sin x$ and $v = e^x$.

$$I = e^x \cos x + \int e^x \sin x \, dx.$$

The remaining integral still contains a product. Apply integration by parts again.

Integration by Parts: Second Application

Let $J = \int e^x \sin x \, dx$. Choose $u = \sin x$ and $v' = e^x$. Then $u' = \cos x$ and $v = e^x$.

$$\begin{aligned} J &= e^x \sin x - \int e^x \cos x \, dx \\ &= e^x \sin x - I. \end{aligned}$$

Substitute this into the first equation:

$$\begin{aligned} I &= e^x \cos x + J \\ &= e^x \cos x + e^x \sin x - I. \end{aligned}$$

Integration by Parts: Solving for I

The original integral appears on both sides:

$$\begin{aligned} I &= e^x(\cos x + \sin x) - I, \\ 2I &= e^x(\cos x + \sin x), \\ I &= \frac{e^x}{2}(\cos x + \sin x) + C. \end{aligned}$$

Differentiating the result gives $e^x \cos x$.

Partial Fractions

Decompose

$$\frac{1}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}.$$

Multiply through by $(x-1)(x-2)$:

$$1 = A(x-2) + B(x-1).$$

Substitute values that make one term zero:

$$x = 1: \quad 1 = -A \quad \Rightarrow \quad A = -1,$$

$$x = 2: \quad 1 = B.$$

Therefore,

$$\frac{1}{(x-1)(x-2)} = -\frac{1}{x-1} + \frac{1}{x-2}.$$

Partial Fractions: Example 1

Evaluate

$$\int \frac{dx}{(x-1)(x-2)}.$$

Substitute the partial fractions found on the previous slide:

$$\begin{aligned}\int \frac{dx}{(x-1)(x-2)} &= \int \left(-\frac{1}{x-1} + \frac{1}{x-2} \right) dx \\ &= -\ln|x-1| + \ln|x-2| + C.\end{aligned}$$

Partial Fractions: Example 2

Evaluate

$$\int \frac{3x + 5}{(x + 1)(x + 2)} dx.$$

Set

$$\frac{3x + 5}{(x + 1)(x + 2)} = \frac{A}{x + 1} + \frac{B}{x + 2}.$$

Then $3x + 5 = A(x + 2) + B(x + 1)$. Substitution gives

$$x = -1 : \quad 2 = A,$$

$$x = -2 : \quad -1 = -B \quad \Rightarrow \quad B = 1.$$

Partial Fractions: Example 2

Continued

The decomposition is

$$\frac{3x + 5}{(x + 1)(x + 2)} = \frac{2}{x + 1} + \frac{1}{x + 2}.$$

Integrate each term:

$$\begin{aligned}\int \frac{3x + 5}{(x + 1)(x + 2)} dx &= 2 \int \frac{dx}{x + 1} + \int \frac{dx}{x + 2} \\ &= 2 \ln |x + 1| + \ln |x + 2| + C.\end{aligned}$$

Partial Fractions: Example 3

Evaluate

$$\int \frac{2x + 4}{x^2 - 1} dx.$$

Since $x^2 - 1 = (x - 1)(x + 1)$, write

$$\frac{2x + 4}{x^2 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1}.$$

Then $2x + 4 = A(x + 1) + B(x - 1)$. Substitution gives

$$x = 1 : \quad 6 = 2A \quad \Rightarrow \quad A = 3,$$

$$x = -1 : \quad 2 = -2B \quad \Rightarrow \quad B = -1.$$

$$\int \frac{2x + 4}{x^2 - 1} dx = 3 \ln |x - 1| - \ln |x + 1| + C.$$

Repeated Linear Factors

A repeated factor needs one term for each power.

$$\frac{P(x)}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{D}{x-b}.$$

The terms integrate differently:

$$\int \frac{dx}{x-a} = \ln |x-a| + C,$$
$$\int \frac{dx}{(x-a)^2} = -\frac{1}{x-a} + C.$$

Partial Fractions: Example 4

Evaluate

$$\int \frac{x+3}{x(x+1)^2} dx.$$

Write

$$\frac{x+3}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{D}{(x+1)^2}.$$

Multiplying through gives

$$x+3 = A(x+1)^2 + Bx(x+1) + Dx.$$

At $x = 0$, $A = 3$. At $x = -1$, $D = -2$. Comparing x^2 coefficients gives $B = -3$.

Partial Fractions: Example 4

Continued

Therefore,

$$\frac{x+3}{x(x+1)^2} = \frac{3}{x} - \frac{3}{x+1} - \frac{2}{(x+1)^2}.$$

Integrating term by term,

$$\int \frac{x+3}{x(x+1)^2} dx = 3 \ln |x| - 3 \ln |x+1| + \frac{2}{x+1} + C.$$

Irreducible Quadratic Factors

A quadratic that does not factor over the real numbers needs a linear numerator:

$$\frac{P(x)}{(x - c)(x^2 + a^2)} = \frac{A}{x - c} + \frac{Bx + D}{x^2 + a^2}.$$

Split the quadratic term after finding the constants. A numerator proportional to $2x$ gives a logarithm. A constant numerator gives an inverse tangent.

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C, \quad a > 0.$$

Partial Fractions: Example 5

Evaluate

$$\int \frac{dx}{x(x^2 + 1)}.$$

Write

$$\frac{1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + D}{x^2 + 1}.$$

Multiplying through gives

$$1 = A(x^2 + 1) + (Bx + D)x.$$

Comparing coefficients gives $A = 1$, $B = -1$, and $D = 0$.

Partial Fractions: Example 5

Continued

The integral becomes

$$\begin{aligned}\int \frac{dx}{x(x^2 + 1)} &= \int \frac{dx}{x} - \int \frac{x}{x^2 + 1} dx \\ &= \ln |x| - \frac{1}{2} \ln(x^2 + 1) + C.\end{aligned}$$

The second term uses $u = x^2 + 1$, so $du = 2x dx$.

Polynomial Division First

If the numerator's degree is at least the denominator's degree, divide before using partial fractions.

$$\frac{x^2 + 3x + 5}{x + 1} = x + 2 + \frac{3}{x + 1}.$$

Hence

$$\begin{aligned}\int \frac{x^2 + 3x + 5}{x + 1} dx &= \int \left(x + 2 + \frac{3}{x + 1} \right) dx \\ &= \frac{x^2}{2} + 2x + 3 \ln |x + 1| + C.\end{aligned}$$

Partial Fractions: Practice

Evaluate each integral.

1. $\int \frac{5x + 1}{x(x + 1)} dx$

2. $\int \frac{dx}{x(x + 2)^2}$

3. $\int \frac{dx}{x(x^2 + 4)}$

Partial Fractions: Practice Answers

1. $\ln|x| + 4\ln|x+1| + C$

2. $\frac{1}{4}\ln|x| - \frac{1}{4}\ln|x+2| + \frac{1}{2(x+2)} + C$

3. $\frac{1}{4}\ln|x| - \frac{1}{8}\ln(x^2+4) + C$

Integrating Powers of Cosine

Write

$$I_n = \int \cos^n x \, dx, \quad n \in \{0, 1, 2, \dots\}.$$

The first two cases are

$$I_0 = \int 1 \, dx = x + C,$$

$$I_1 = \int \cos x \, dx = \sin x + C.$$

Integration by parts relates I_n to I_{n-2} .

A Reduction Formula for $\cos^n x$

Write $I_n = \int \cos^{n-1} x \cos x \, dx$. Choose $u = \cos^{n-1} x$ and $v' = \cos x$. Then

$$u' = -(n-1) \cos^{n-2} x \sin x, \quad v = \sin x.$$

Integration by parts gives

$$\begin{aligned} I_n &= \sin x \cos^{n-1} x + (n-1) \int \sin^2 x \cos^{n-2} x \, dx \\ &= \sin x \cos^{n-1} x + (n-1)(I_{n-2} - I_n). \end{aligned}$$

Rearranging, for $n \geq 2$,

$$I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}.$$

An Even Power: $\cos^4 x$

First use $n = 2$:

$$I_2 = \frac{1}{2} \sin x \cos x + \frac{1}{2}x + C.$$

Now use $n = 4$:

$$\begin{aligned} I_4 &= \frac{1}{4} \sin x \cos^3 x + \frac{3}{4} I_2 \\ &= \frac{1}{4} \sin x \cos^3 x + \frac{3}{8} \sin x \cos x + \frac{3}{8}x + C. \end{aligned}$$

An Odd Power: $\cos^5 x$

For odd powers, a substitution is often shorter. Write

$$\cos^5 x = \cos x(1 - \sin^2 x)^2.$$

Let $u = \sin x$, so $du = \cos x \, dx$.

$$\begin{aligned}\int \cos^5 x \, dx &= \int (1 - u^2)^2 \, du \\ &= u - \frac{2}{3}u^3 + \frac{1}{5}u^5 + C \\ &= \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C.\end{aligned}$$

Calculating $\int \cos^n x \, dx$

For a nonnegative integer n , reduce the power by two until reaching I_0 or I_1 :

$$\boxed{I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}}, \quad n \geq 2.$$

Use the starting values

$$I_0 = x, \quad I_1 = \sin x,$$

and add one constant of integration to the final answer.